Asymptotic Methods

Part II

Recorded from the lecturer in 2026-2027.

Pre-requisites

- Complex methods / analysis.
- Refer to Section II in Stuart.

1. Intro & Motivation

- Most real problems don't have closed form solutions, or if they do, they are unworkable or difficult to interpret.
- Numerical approximations can be useful, but they can be expensive, they may have bad numerical properties, or solutions may be inscrutable.
- Taylor series, etc. can be limited in applicability.
- Asymptotic methods emphasise usefulness, interpretability as well as accuracy.

Examples

e.g. 1)

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \qquad \mathrm{Re}(z) > 0.$$

For n an integer,

$$\Gamma(n+1) = n! \sim \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n} \left(1 + \frac{1}{12n} + \frac{1}{288n^2} + \cdots\right) \qquad \text{as } n \to \infty.$$

e.g. 2)

$$x^2 y'' + xy' - x^2 y = 0.$$

Exact series solution:

$$I_0(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{2^{2n}(n!)^2}.$$

Liouville–Green:

$$\implies I_0(x) \sim (2\pi x)^{-1/2} e^x \qquad \text{as } x \to \infty.$$

e.g. 3)

$$y'' + k^2 e^{-x^2/2} y = 0,$$

with

$$y \to y_0 \qquad \text{as } x \to \infty,$$

$$y \to \pm y_0 \qquad \text{as } x \to -\infty.$$

$$k = ?, \qquad k = k_0, k_1, k_2, \ldots$$

$$\implies k_n \sim \frac{\sqrt{\pi}}{2}\left(n + \frac{1}{2}\right) \qquad \text{as } n \to \infty.$$

WKBJ.

- Convergent methods typically give arbitrarily accurate approximations with more & more terms/iterations.
- Asymptotic methods give arbitrarily accurate approximations as some variable approaches a limit. Thus they always come with a qualifier, e.g. $z \to 0$ or $n \to \infty$.

2. Asymptotic Expansions

2.1 Definitions & Notation

(a) Asymptotic Equality/Equivalence

For 2 functions $f(x)$ and $g(x)$, we have

$$f(x) \sim g(x) \qquad \text{about } x = x_0$$

if

$$\boxed{\lim_{x \to x_0} \frac{f(x)}{g(x)} = 1.}$$

Remarks:

- Usually we want to approximate f with g.
- We can write
$$f(x) = \underbrace{g(x)}_{\text{dominant}} + \underbrace{[f(x) - g(x)]}_{\text{subdominant}}.$$
- Note $\sim$ is an equivalence relation (that is reflexive, transitive, symmetric).
- It respects products:
$$f_1 \sim g_1, \qquad f_2 \sim g_2$$
$$\implies f_1 f_2 \sim g_1 g_2.$$
- It is linear:
$$\alpha f_1 + \beta f_2 \sim \alpha g_1 + \beta g_2,$$
where α, β are constants.
- Definition has problems when dealing with oscillatory functions that go through 0 as $x \to x_0$.

(b) "Little o"

Suppose f & g are real functions.

We write

$$\boxed{f(x) = o(g(x)) \qquad \text{as } x \to x_0}$$

if

$$\boxed{\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0.}$$

Remark:

$$f(x) = o(g(x))$$

can also be written as

$$f(x) \ll g(x) \qquad \text{as } x \to x_0.$$

- Can be framed in inequalities, e.g. $x_0 = \infty$ and let $g > 0$.
  Then $f = o(g)$ as $x \to \infty$ is equivalent to

$$\forall \varepsilon > 0, \quad \exists a \quad \text{s.t. when } x > a,$$

$$\boxed{|f| \leq \varepsilon g.}$$

(c) "Big O"

We write

$$f(x) = O(g(x)) \qquad \text{as } x \to x_0$$

for $g > 0$, if

$$\exists C > 0 \quad \text{s.t.}$$

$$\boxed{|f(x)| \leq C g(x)}$$

for x sufficiently close to x_0.

Warning:

In some areas of physics people use big-O and $\sim$ interchangeably.

Little Exercises: Show That

(a)

$$f(x) \sim g(x) \iff f - g = o(g) \qquad \text{as } x \to x_0.$$

(b)

$$\sin x \ll x^2 \qquad \text{as } x \to \infty.$$

(c)

$$12x^5 + 6x^3 + x = O(x^5) \qquad \text{as } x \to \infty.$$

(d)

$$e^{ax} \ll e^{bx} \qquad \text{as } x \to \infty,$$

if $b > a > 0$.

(e)

$$e^x \gg x^N \qquad \text{as } x \to \infty,$$

$$\forall N \in \mathbb{Z}^+.$$

2.2 Asymptotic Expansions

- Have unworkable/unknown $f(x)$.
- Want to know its behaviour in a limit.
- Want to find a simpler/known $g(x)$ so that

$$f \sim g.$$

- Also want to break up g into a hierarchy of approximations, s.t.

$$g(x) = \sum_{k=0}^{N} a_k \phi_k(x),$$

where a_k are coefficients, ϕ_k are known & workable, and where adding more terms gives a better approximation that can be estimated.

(a) Asymptotic Sequence

A sequence of functions

$$\{\phi_0(x), \phi_1(x), \ldots\}$$

is an "asymptotic sequence" as $x \to x_0$ if

$$\boxed{\phi_{k+1}(x) \ll \phi_k(x) \qquad \text{as } x \to x_0, \quad \forall k.}$$

e.g. 1)

$$\phi_k = (x-a)^k \qquad \text{around } x = a.$$

Check:

$$\frac{\phi_{k+1}}{\phi_k} = x - a \to 0 \qquad \text{as } x \to a. \quad \checkmark$$

e.g. 2)

$$\phi_k = e^{-kx} x^k \qquad \text{as } x \to \infty \text{ or } x \to -\infty.$$

Check:

$$\frac{\phi_{k+1}}{\phi_k} = e^{-x} x.$$

$$e^{-x} x \to 0 \quad \text{as } x \to \infty \qquad \checkmark$$
$$e^{-x} x \to \infty \quad \text{as } x \to -\infty \qquad \times$$

e.g. 3)

$$\phi_k = \cos(nx) \qquad \text{as } x \to \infty.$$

$\times$

e.g. 4)

$$\{|x|, \log|x|, \log|\log|x||, \ldots\} \qquad \text{as } x \to \infty.$$

?

(b) Asymptotic Expansion

- Suppose $f(x)$ is a function we want to understand near $x = x_0$.
- Let $\{\phi_k\}_{k=0}^{\infty}$ be an asymptotic sequence.

Then, the finite sum

$$\sum_{k=0}^{N} a_k \phi_k(x)$$

for some coefficients, is an (N-term) asymptotic approximation of $f(x)$ as $x \to x_0$ if:

$$\boxed{f(x) = \sum_{k=0}^{N} a_k \phi_k(x) + o(\phi_N(x))} \qquad \text{as } x \to x_0.$$

Alternatively, we can write the above as

$$\boxed{\frac{f(x) - \sum_{k=0}^{N} a_k \phi_k(x)}{\phi_N(x)} \longrightarrow 0} \qquad \text{as } x \to x_0.$$

End of lecture 1

(c) Asymptotic expansions/series

If (b) holds for *all* N, then this is called an "asymptotic series or expansion", and we write

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k(x) \qquad \text{as } x \to x_0.$$

Attention: No claim of convergence is made. It is just notation. We never fix x and then take $N \to \infty$.

Instead, we have an infinite number of approximations (of different N), and $x \to x_0$ is taken in each separately!

(d) Properties of asymptotic expansions and remarks

(I) Mathematical operations

Suppose $\{\phi_k(x)\}$ is an asymptotic sequence as $x \to x_0$, and let

$$f \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad g \sim \sum_{k=0}^{\infty} b_k \phi_k.$$

Linearity

$$\alpha f + \beta g \sim \sum_{k=0}^{\infty} (\alpha a_k + \beta b_k) \phi_k \qquad \text{as } x \to x_0.$$

Multiplication

If $\phi_k = (x - x_0)^k$, then

$$fg \sim \sum_{k=0}^{\infty} c_k \phi_k, \qquad c_k = \sum_{j=0}^{k} a_j b_{k-j}.$$

Division

See 3.8 BO.

Integration

Term-by-term integration is usually okay.

For example, if $\phi_k = (x - x_0)^k$, then

$$\int_{x_0}^{x} f(t)\,dt \sim \sum_{k=0}^{\infty} \frac{a_k (x - x_0)^{k+1}}{k+1} \qquad \text{as } x \to x_0.$$

Exercise: Use the definitions to prove this.

Differentiation

Term-by-term differentiation is not always okay.

To do this, you need $f'(x)$ to be continuous and to have an asymptotic series, i.e.

$$f'(x) \sim \sum_{k=0}^{\infty} c_k \phi_k(x).$$

(II) Different asymptotic representations

Suppose $\{\phi_k\}$ and $\{\psi_k\}$ are different asymptotic sequences as $x \to x_0$. Also,

$$f \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad f \sim \sum_{k=0}^{\infty} b_k \psi_k \qquad \text{as } x \to x_0.$$

Then a_k and b_k are not the same.

For example,

$$f(x) = 1 + \frac{1}{x}.$$

Consider $\phi_k = x^{-k}$ as $x \to \infty$. Then

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad a_0 = 1, \quad a_1 = 1, \quad a_n = 0 \quad (n > 1).$$

Now consider

$$\psi_k = (1 + x)^{-k}.$$

Then

$$\begin{aligned} f = 1 + \frac{1}{x} &= \frac{1}{1 - (x+1)^{-1}} \\ &= 1 + \frac{1}{1+x} + \frac{1}{(1+x)^2} + \cdots \\ &\sim \sum_{k=0}^{\infty} b_k \psi_k, \end{aligned}$$

where

$$b_k = 1 \qquad \forall k.$$

(III) Uniqueness of coefficients

Let $\{\phi_k\}$ be an asymptotic sequence, and

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k \qquad \text{as } x \to x_0.$$

Then the coefficients a_k are unique.

In fact, the a_k can be constructed:

$$a_0 = \lim_{x \to x_0} \frac{f(x)}{\phi_0(x)},$$

$$a_n = \lim_{x \to x_0} \left[\frac{f(x) - \sum_{k=0}^{n-1} a_k \phi_k(x)}{\phi_n(x)} \right].$$

Proof. First, we know

$$\lim_{x \to x_0} \frac{f(x) - \sum_{k=0}^{N} a_k \phi_k}{\phi_N} = 0 \qquad \forall N.$$

Let $N = 0$. Then

$$\lim_{x \to x_0} \frac{f(x) - a_0 \phi_0}{\phi_0} = 0,$$

$$\lim_{x \to x_0} \left(\frac{f(x)}{\phi_0} \right) - a_0 = 0.$$

Use induction on N.

Suppose the formulae are true for $N = n$, and the coefficients are unique for $k \leq n$. Then

$$\lim_{x \to x_0} \frac{f(x) - \sum_{k=0}^{n} a_k \phi_k}{\phi_n} = 0$$

$$= \lim_{x \to x_0} \left(\frac{f(x) - \sum_{k=0}^{n-1} a_k \phi_k}{\phi_n} \right) - a_n.$$

Alternative proof: see p. 3 in Manton.

(IV) An asymptotic series does not uniquely determine the function it is describing

For example, suppose

$$f(x) \sim g(x) \qquad \text{as } x \to x_0,$$

and

$$h(x) \ll f(x) \qquad \text{as } x \to x_0.$$

Then

$$f(x) + h(x) \sim g(x) \qquad \text{as } x \to x_0.$$

So, if g is an asymptotic expansion, it can describe both f and $f+h$.

For example,

$$f_1 = \frac{1}{1+x}, \qquad f_2 = \frac{1+e^{-x}}{1+x}.$$

Both are

$$\sim \sum_{k=0}^{\infty} (-1)^k x^{-(k+1)} \qquad \text{as } x \to \infty.$$

(V) Power series

In this course, we mostly take

$$\phi_k = (x - x_0)^k \qquad \text{as } x \to x_0,$$

or

$$\phi_k = x^{-k} \qquad \text{as } x \to \infty.$$

In fact, a Taylor series about a point is an asymptotic series about that point:

$$f(x) - \sum_{k=0}^{N} \frac{f^{(k)}(a)(x-a)^k}{k!} = o\big(|x-a|^N\big)\,.$$

Exercise: Prove this using the definitions and the integral form of the remainder term.

Terminology: A term is said to be "small beyond all orders" in some limit if it cannot be represented by any power series or, equivalently, is smaller than any power.

For example, e^{-x} is small beyond all orders in inverse powers of x as $x \to \infty$:

$$e^{-x} \ll x^{-k} \qquad \forall k, \qquad \text{as } x \to \infty.$$

(VI) Optimal truncation

Question: If an asymptotic series diverges, how many terms should I include in the sum?

If

$$f(x) \sim \sum_{n=0}^{\infty} a_n \psi_n(x) \qquad \text{as } x \to x_0,$$

and we want to get the best approximation to $f(x)$ at $x = x_1$, where x_1 is close to x_0, then truncate the sum at the smallest term (or the one before).

So stop adding terms when

$$|a_k \psi_k(x_1)| < |a_{k+1} \psi_{k+1}(x_1)|.$$

This gives you an unexpectedly excellent approximation.

End of lecture 2

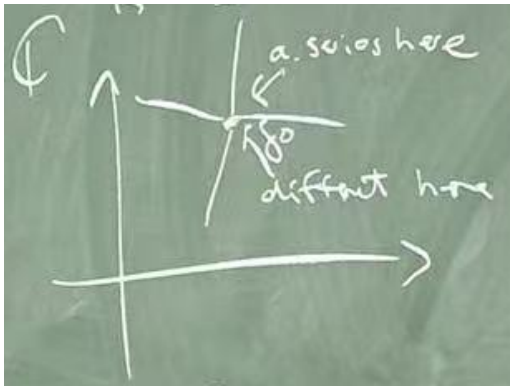
2.3 A. expansions in $\mathbb{C}$ — Stokes phenomenon

- So far our defns relied on real limits, $x, x_0 \in \mathbb{R}$ etc.
- When we use complex variables, $x = z \in \mathbb{C}$ and $z \to z_0$, typically z must be restricted to a sector of $\mathbb{C}$ around z_0 for some of our definitions to work.

  [See III.2 in Stuart]

  e.g. little o & O defns need to bring in moduli of functions.

- Even if $f(z)$ analytic at z_0, it may have different a. series as z_0 is approached from different sectors in $\mathbb{C}$. This is Stokes phenomenon.



e.g.

$$f(z) = \sinh(z^2) = \frac{1}{2}\left(e^{z^2} - e^{-z^2}\right), \qquad z \in \mathbb{C}.$$

What is its behaviour as $|z| \to \infty$?

Which of e^{z^2} or e^{-z^2} is dominant?

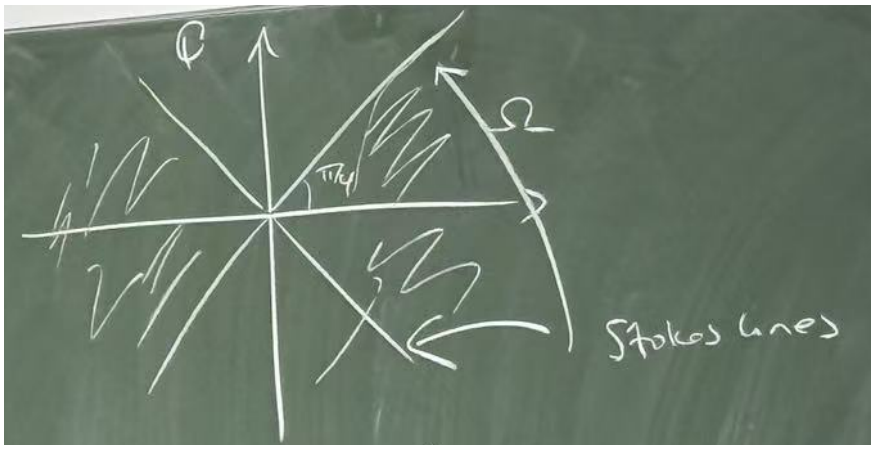
$$\left|\frac{e^{-z^2}}{e^{z^2}}\right| = \left|e^{-2z^2}\right| = e^{-2\,\mathrm{Re}(z^2)}.$$

(a) Suppose $|z| \to \infty$ & $\mathrm{Re}(z^2) > 0$.

$$\text{ratio} \to 0 \quad \Rightarrow \quad e^{-z^2} \ll e^{z^2} \quad \Rightarrow \quad f(z) \sim \frac{1}{2}e^{z^2} \quad \text{in this limit.}$$

$$\mathrm{Re}(z^2) > 0 \quad \Rightarrow \quad \Omega: \quad -\frac{\pi}{4} + \delta < \arg z < \frac{\pi}{4} - \delta, \quad \text{and} \quad \frac{3\pi}{4} + \delta < \arg z < \frac{5\pi}{4} - \delta,$$

for small δ.



(b) $|z| \to \infty$ & $\mathrm{Re}(z^2) < 0$.

$$\text{ratio} \to \infty \quad \Rightarrow \quad e^{z^2} \ll e^{-z^2} \quad \& \quad f \sim -\frac{1}{2}e^{-z^2}.$$

$$\widetilde{\Omega}: \quad \frac{\pi}{4}+\delta < \arg z < \frac{3\pi}{4}-\delta \quad \& \quad -\frac{3\pi}{4}+\delta < \arg z < -\frac{\pi}{4}-\delta.$$

$$f(z) \sim \begin{cases} \frac{1}{2}e^{z^2}, & |z|\to\infty \ \& \ z\in\Omega, \\ -\frac{1}{2}e^{-z^2}, & |z|\to\infty \ \& \ z\in\widetilde{\Omega}. \end{cases}$$

- A. behaviour of f as $|z|\to\infty$ depends on which wedge $z\to\infty$ within.
- $\mathbb{C}$ is divided into sectors, separated by special lines, called Stokes lines. [Warning, sometimes called anti-Stokes lines!!]

3. Approximations of Integrals

3.1 Taylor expanding the integrand — Stieltjes integrals

(Warm up problem)

Consider:

$$I(x) = \int_0^\infty \frac{e^{-t}}{1+xt}\,dt, \qquad x\geq 0.$$

- We want to understand I's behaviour when $x\to 0^+$.
- How about we expand $\dfrac{1}{1+xt}$?

$$\frac{1}{1+xt} = \sum_{n=0}^{\infty}(-1)^n x^n t^n \qquad \times \quad t \text{ can be big!}$$

Consider instead

$$\sum_{n=0}^{N}(-xt)^n = \frac{1-(-xt)^{N+1}}{1+xt} = \frac{1}{1+xt} - \underbrace{\frac{(-xt)^{N+1}}{1+xt}}_{r_N}.$$

$$\Rightarrow \quad \frac{1}{1+xt} = \sum_{n=0}^{N}(-xt)^n + r_N.$$

$$\begin{aligned} I(x) &= \int_0^\infty e^{-t}\left[\sum_{n=0}^{N}(-1)^n x^n t^n + r_N\right]dt \\ &= \sum_{n=0}^{N}(-1)^n x^n \underbrace{\int_0^\infty t^n e^{-t}\,dt}_{=n!} + R_N \\ &= \sum_{n=0}^{N}(-1)^n n!\, x^n + R_N \\ &= \sum c_n \phi_n(x) + R_N. \end{aligned}$$

$\phi_n = x^n$ as $x\to 0^+$, so is a. sequence!

Need to show $R_N = o(x^N)$ as $x\to 0^+$.

$$\begin{aligned} \frac{R_N}{x^N} &= \frac{x^{N+1}}{x^N}\int_0^\infty \frac{t^{N+1}e^{-t}}{1+xt}\,dt \\ &\leq x\int_0^\infty t^{N+1}e^{-t}\,dt = x(N+1)! \longrightarrow 0 \quad \text{as } x\to 0^+. \end{aligned}$$

Note $1 + xt > 1$. True $\forall N$.

$$\Rightarrow \quad I(x) \sim \sum_{n=0}^{\infty} (-1)^n n!\, x^n \qquad \text{as } x \to 0^+.$$

3.2 Integration by parts

Recap:

$$\int_a^b f'(t)g(t)\,dt = [fg]_a^b - \int_a^b fg'\,dt.$$

Generalisation:

$$\begin{aligned} I &= \int_a^b f(t)g^{(n)}(t)\,dt \\ &= \sum_{k=1}^{n} (-1)^{k-1} \left[f^{(k-1)}(t)g^{(n-k)}(t) \right]_a^b + (-1)^n \int_a^b f^{(n)}(t)g(t)\,dt. \end{aligned}$$

- Repeated integration by parts can generate an a. approximation under certain circumstances.

e.g. How does

$$F(x) = \int_x^\infty e^{-t^2/2}\,dt$$

behave as $x \to \infty$?

$$\begin{aligned} F(x) &= \int_x^\infty \frac{te^{-t^2/2}}{t}\,dt \\ &= \left[-\frac{1}{t} e^{-t^2/2} \right]_x^\infty + \int_x^\infty \frac{e^{-t^2/2}}{t^2}\,dt \\ &= \underbrace{\frac{e^{-x^2/2}}{x}}_{\text{a. approximation}} + \underbrace{\int_x^\infty \frac{e^{-t^2/2}}{t^2}\,dt}_{\text{error}}. \end{aligned}$$

Need to check that

$$\text{error} = o\left(\frac{e^{-x^2/2}}{x} \right) \qquad \text{as } x \to \infty.$$

$$\begin{aligned} \text{error} &= \int_x^\infty \frac{te^{-t^2/2}}{t^3}\,dt \qquad (\text{note } t \geq x) \\ &\leq \frac{1}{x^3} \int_x^\infty te^{-t^2/2}\,dt = \frac{1}{x^3} e^{-x^2/2} \\ &= o\left(\frac{e^{-x^2/2}}{x} \right) \qquad \text{as } x \to \infty. \end{aligned}$$

$$\Rightarrow \quad F(x) \sim \frac{e^{-x^2/2}}{x} \qquad \text{as } x \to \infty.$$

In fact we find

$$F(x) \sim \frac{e^{-x^2/2}}{x} \sum_{n=0}^{\infty} (-1)^n \frac{(2n-1)!!}{x^{2n}} \qquad \text{as } x \to \infty.$$

$$(2n-1)!! = (2n-1)(2n-3)(2n-5)\cdots 3 \cdot 1.$$

Exercise: Show this.

End of lecture 3

Example 2. Find the behaviour of

$$I(x) = \int_a^b f(t)e^{xt}\,dt \qquad \text{as } x \to \infty,$$

where $b > a$, $f(b) \neq 0$, and f is smooth on $[a, b]$.

As $x \to \infty$, the integral is dominated by a narrow region near $t = b$ because of the exponential. Very roughly,

$$I(x) \approx f(b) \int_a^b e^{xt}\,dt = \frac{f(b)}{x}(e^{xb} - e^{xa}) \sim \frac{f(b)e^{xb}}{x},$$

using $e^{xb} \gg e^{xa}$ as $x \to \infty$.

Repeated integration by parts, using $e^{xt} = \dfrac{d}{dt}\left(\dfrac{e^{xt}}{x}\right)$, gives

$$\begin{aligned} I(x) &= \sum_{k=1}^{N}(-1)^{k-1}\left[f^{(k-1)}(t)e^{xt}x^{-k}\right]_a^b + \frac{(-1)^N}{x^N}\int_a^b f^{(N)}(t)e^{xt}\,dt \\ &= e^{xb}\big(J_1 - J_2 + J_3\big), \end{aligned}$$

where

$$\begin{aligned} J_1 &= \sum_{k=1}^{N}(-1)^{k-1}f^{(k-1)}(b)x^{-k}, \\ J_2 &= e^{-x(b-a)}\sum_{k=1}^{N}(-1)^{k-1}f^{(k-1)}(a)x^{-k}, \\ J_3 &= \frac{(-1)^N}{x^N}\int_a^b f^{(N)}(t)e^{x(t-b)}\,dt. \end{aligned}$$

J_1 could be an N-term a. approximation with $\phi_k = x^{-k}$ as $x \to \infty$, but we need to show that

$$J_2 = o(x^{-N}), \qquad J_3 = o(x^{-N}) \qquad \text{as } x \to \infty.$$

For the first term,

$$\frac{J_2}{x^{-N}} = e^{-x(b-a)}\sum_{k=1}^{N}(-1)^{k-1}f^{(k-1)}(a)x^{N-k} \longrightarrow 0 \qquad \text{as } x \to \infty,$$

which is true for every N. For the remainder,

$$\begin{aligned} \left|\frac{J_3}{x^{-N}}\right| &\le \int_a^b |f^{(N)}(t)|e^{x(t-b)}\,dt \\ &\le \max_{a\le t\le b}|f^{(N)}(t)|\int_a^b e^{x(t-b)}\,dt \\ &= \max_{a\le t\le b}|f^{(N)}(t)|\frac{1-e^{-x(b-a)}}{x} \longrightarrow 0 \qquad \text{as } x \to \infty. \end{aligned}$$

This holds for every N for which f has N continuous derivatives. If f is infinitely differentiable, then

$$I(x) = \int_a^b f(t)e^{xt}\,dt \sim \sum_{k=1}^{\infty}(-1)^{k-1}\frac{f^{(k-1)}(b)e^{xb}}{x^k} \qquad \text{as } x \to \infty.$$

3.3 The Gamma function

Recall

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \qquad \mathrm{Re}(z) > 0.$$

(a) Γ is analytic for $\mathrm{Re}(z) > 0$.

(b) $\Gamma(z+1) = z\Gamma(z)$ (recursion relation).

(c) $\Gamma(1) = 1$ and $\Gamma(n+1) = n!$ for positive integers n.

(d) For $x > 0$ and $\mathrm{Re}(z) > 0$,

$$\int_0^\infty t^{z-1} e^{-xt}\, dt = \frac{\Gamma(z)}{x^z}.$$

(e) Generalisation of binomial coefficients:

$$\binom{z}{k} = \frac{1}{k!}\frac{\Gamma(z+1)}{\Gamma(z-k+1)}, \qquad z \in \mathbb{C}, \quad k = 0, 1, 2, 3, \ldots,$$

with the quotient interpreted by continuation where necessary.

$$(1+t)^z = \sum_{k=0}^{\infty} \binom{z}{k} t^k, \qquad |t| < 1.$$

(f) Reflection and duplication formulae: see Section 3.2 in Chapling.

3.4 Watson's lemma

Consider

$$F(x) = \int_0^T f(t) e^{-xt}\, dt \qquad \text{as } x \to \infty, \qquad 0 < T \leq \infty.$$

Suppose

$$f(t) \sim t^\alpha \sum_{n=0}^{\infty} a_n t^{\beta n} \qquad \text{as } t \to 0^+, \qquad \alpha > -1, \quad \beta > 0.$$

Also assume that either

(i) $|f(t)| < Ke^{bt}$ for all $t \in [0, T]$, for some $K, b > 0$; or

(ii) $\displaystyle\int_0^T |f(t)|\, dt < \infty$.

Then

$$\boxed{F(x) \sim \sum_{n=0}^{\infty} a_n \frac{\Gamma(\alpha + \beta n + 1)}{x^{\alpha+\beta n+1}}} \qquad \text{as } x \to \infty.$$

This is *Watson's lemma*.

Example 2.

$$xy'' + (a + 1 - x)y' - ay = 0, \qquad a > 0.$$

One solution is

$$M(x) = \frac{e^x}{\Gamma(a)} \int_0^1 e^{-xt}(1-t)^{a-1}\, dt.$$

What is its behaviour as $x \to \infty$?

Here $f(t) = (1-t)^{a-1}$ is integrable, so case (ii) applies, and

$$f(t) = (1-t)^{a-1} = \sum_{n=0}^{\infty} (-1)^n \frac{\Gamma(a)}{\Gamma(a-n)n!}\, t^n, \qquad |t| < 1.$$

Thus $\alpha = 0$, $\beta = 1$, and Watson's lemma gives

$$\begin{aligned} M(x) &\sim \frac{e^x}{\Gamma(a)} \sum_{n=0}^{\infty} \frac{(-1)^n \Gamma(a)\Gamma(n+1)}{\Gamma(a-n)n!} \frac{1}{x^{n+1}} \\ &= e^x \sum_{n=0}^{\infty} \frac{(-1)^n}{\Gamma(a-n)x^{n+1}} \qquad \text{as } x \to \infty. \end{aligned}$$

In particular,

$$M(x) \sim \frac{e^x}{x\Gamma(a)} \qquad \text{as } x \to \infty.$$

Proof of Watson's lemma. Write

$$\begin{aligned} F(x) &= \int_0^{\epsilon} f(t) e^{-xt}\, dt + \int_{\epsilon}^{T} f(t) e^{-xt}\, dt \\ &\equiv F_1(x) + F_2(x), \end{aligned}$$

where $\epsilon > 0$ is small and fixed, with $\epsilon < T$.

First, show that F_2 is small.

Case (i): for $x > b$,

$$\begin{aligned} |F_2(x)| &\le \int_{\epsilon}^{T} K e^{bt} e^{-xt}\, dt \\ &\le \int_{\epsilon}^{\infty} K e^{-(x-b)t}\, dt \\ &= \frac{K}{x-b} e^{(b-x)\epsilon} \\ &= O\left(\frac{e^{-\epsilon x}}{x}\right) \qquad \text{as } x \to \infty. \end{aligned}$$

Case (ii):

$$\begin{aligned} |F_2(x)| &\le \int_{\epsilon}^{T} |f(t)| e^{-xt}\, dt \\ &\le e^{-\epsilon x} \int_{\epsilon}^{T} |f(t)|\, dt \\ &\le e^{-\epsilon x} \int_{0}^{T} |f(t)|\, dt \\ &= O(e^{-\epsilon x}) \qquad \text{as } x \to \infty. \end{aligned}$$

In both cases, F_2 is small beyond all orders in inverse powers of x.

End of lecture 4

3.3 Gamma function (again)

Γ can be analytically continued to $\mathbb{C}$, except at $z = 0, -1, -2, \ldots$, using

$$\Gamma(z) = \frac{\Gamma(z+1)}{z}.$$

Proof of Watson's lemma (continued). Recall that

$$F(x) = \int_0^\epsilon f(t)e^{-xt}\,dt + \int_\epsilon^T f(t)e^{-xt}\,dt = F_1(x) + F_2(x),$$

with $F_2(x) = O(e^{-\epsilon x})$ as $x \to \infty$. We want to show, for every N, that

$$F_1(x) = \sum_{n=0}^{N} a_n \frac{\Gamma(\alpha+\beta n+1)}{x^{\alpha+\beta n+1}} + o\Big(x^{-\alpha-\beta N-1}\Big).$$

Using property (d) of the Gamma function, consider

$$\begin{aligned}
&\left|F_1(x) - \sum_{n=0}^{N} a_n \frac{\Gamma(\alpha+\beta n+1)}{x^{\alpha+\beta n+1}}\right| \\
&= \left|\int_0^\epsilon f(t)e^{-xt}\,dt - \int_0^\infty \sum_{n=0}^{N} a_n t^{\alpha+\beta n} e^{-xt}\,dt\right| \\
&\le \int_0^\epsilon e^{-xt}\left|f(t) - \sum_{n=0}^{N} a_n t^{\alpha+\beta n}\right|\,dt + \left|\int_\epsilon^\infty e^{-xt}\sum_{n=0}^{N} a_n t^{\alpha+\beta n}\,dt\right| \\
&\equiv R_1 + R_2.
\end{aligned}$$

For R_2, put $u = t - \epsilon$. Then

$$\begin{aligned}
R_2 &\le e^{-\epsilon x}\sum_{n=0}^{N}|a_n|\int_\epsilon^\infty e^{-x(t-\epsilon)}t^{\alpha+\beta n}\,dt \\
&= e^{-\epsilon x}\sum_{n=0}^{N}|a_n|\int_0^\infty e^{-xu}(\epsilon+u)^{\alpha+\beta n}\,du.
\end{aligned}$$

For each n, there exist $C_n, k_n > 0$ such that

$$(\epsilon+u)^{\alpha+\beta n} < C_n e^{k_n u} \qquad \text{for all } u \in [0,\infty).$$

Hence, for $x > \max_{0\le n\le N} k_n$,

$$\begin{aligned}
R_2 &\le e^{-\epsilon x}\sum_{n=0}^{N} C_n|a_n|\int_0^\infty e^{(k_n-x)u}\,du \\
&= e^{-\epsilon x}\sum_{n=0}^{N}\frac{C_n|a_n|}{x-k_n} = O\left(\frac{e^{-\epsilon x}}{x}\right) \qquad \text{as } x\to\infty.
\end{aligned}$$

In R_1, use the asymptotic expansion of f. In particular,

$$f(t) - \sum_{n=0}^{N} a_n t^{\alpha+\beta n} = o(t^{\alpha+\beta N}) = O(t^{\alpha+\beta(N+1)}) \qquad \text{as } t\to 0^+,$$

where the last bound follows by taking one more term in the expansion. By the definition of O, for sufficiently small ϵ there is $K_3 > 0$ such that

$$\begin{aligned} R_1 &\le K_3 \int_0^\epsilon e^{-xt} t^{\alpha+\beta(N+1)}\, dt \\ &\le K_3 \int_0^\infty e^{-xt} t^{\alpha+\beta(N+1)}\, dt \\ &= K_3 \frac{\Gamma(\alpha+\beta(N+1)+1)}{x^{\alpha+\beta(N+1)+1}}. \end{aligned}$$

Since $\beta > 0$,

$$R_1 = o\Big(x^{-\alpha-\beta N-1}\Big) \qquad \text{as } x \to \infty.$$

This is true for every N, so

$$F_1(x) \sim \sum_{n=0}^{\infty} a_n \frac{\Gamma(\alpha+\beta n+1)}{x^{\alpha+\beta n+1}}.$$

Since F_2 is smaller than every inverse power of x, the same expansion holds for $F = F_1 + F_2$:

$$F(x) \sim \sum_{n=0}^{\infty} a_n \frac{\Gamma(\alpha+\beta n+1)}{x^{\alpha+\beta n+1}} \qquad \text{as } x \to \infty.$$

3.5 Laplace integrals and Laplace's method

Laplace integrals are of the form

$$I(x) = \int_a^b f(t) e^{x\phi(t)}\, dt,$$

where $b > a$, and f and ϕ are sufficiently nice functions. What happens to I as $x \to \infty$? Watson's lemma deals with the special case $\phi(t) = -t$, $a = 0$, $b = T$.

In general, we have Laplace's *principle of localisation*: the asymptotic expansion of I as $x \to \infty$ is determined by contributions from arbitrarily small regions around the maxima of ϕ. Figure 1 illustrates this principle.

(a) Suppose ϕ is monotonic on $[a, b]$. If $\phi'(t) > 0$ and $f(b) \neq 0$, let

$$u = \phi(t), \qquad t = \phi^{-1}(u).$$

Then

$$I(x) = \int_{\phi(a)}^{\phi(b)} \frac{f(t)}{\phi'(t)} e^{xu}\, du, \qquad t = \phi^{-1}(u).$$

Comparing with Example 2 from Section 3.2 gives

$$I(x) \sim \frac{f(b) e^{x\phi(b)}}{x\phi'(b)} \qquad \text{as } x \to \infty.$$

If $\phi'(t) < 0$ and $f(a) \neq 0$, then similarly

$$I(x) \sim \frac{f(a) e^{x\phi(a)}}{x|\phi'(a)|} \qquad \text{as } x \to \infty.$$

(b) Suppose $\phi(t)$ has one interior maximum at $t = c$. Assume

$$\phi'(c) = 0, \qquad \phi''(c) < 0, \qquad f(c) \neq 0.$$

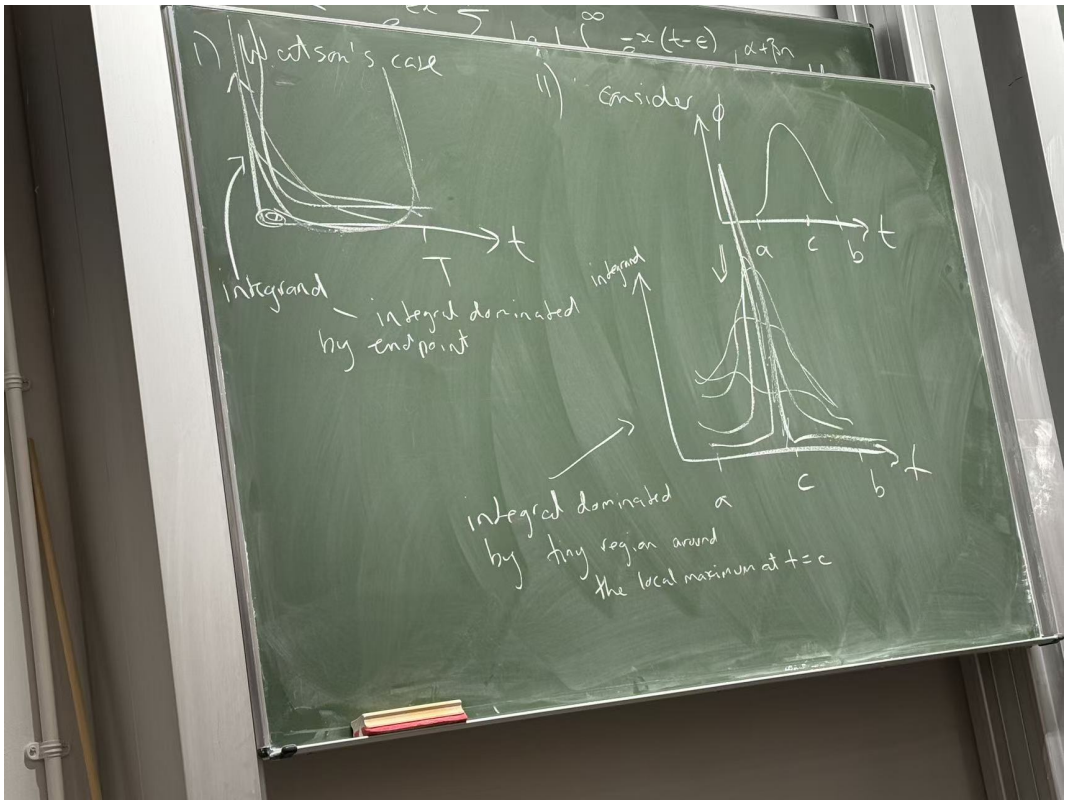


Figure 1: The three blackboard sketches of localisation. Left: Watson's case, where the integral is dominated by an endpoint. Upper right: a phase ϕ with an interior maximum at $t = c$. Lower right: the integrand becomes concentrated in a tiny region around c. Source: lecture blackboard photograph.

For small, positive ϵ, write

$$\Omega_\epsilon = [a, c-\epsilon] \cup [c+\epsilon, b]$$

and

$$\begin{aligned} I(x) &= \int_{c-\epsilon}^{c+\epsilon} f(t) e^{x\phi(t)}\, dt + \int_{\Omega_\epsilon} f(t) e^{x\phi(t)}\, dt \\ &\equiv I_1(x) + I_2(x). \end{aligned}$$

The small region around c is the region responsible for the concentration shown in Figure 1.

First, bound I_2. In the monotonic regions on either side of c,

$$\begin{aligned} I_2(x) &= \int_{c+\epsilon}^{b} f(t) e^{x\phi(t)}\, dt + \int_{a}^{c-\epsilon} f(t) e^{x\phi(t)}\, dt \\ &= O\left(\frac{e^{x\phi(c+\epsilon)}}{x} \right) + O\left(\frac{e^{x\phi(c-\epsilon)}}{x} \right). \end{aligned}$$

Here ϕ is monotonically decreasing on the right and monotonically increasing on the left. But

$$\phi(c+\epsilon) < \phi(c), \qquad \phi(c-\epsilon) < \phi(c),$$

so

$$I_2(x) = o\left(\frac{e^{x\phi(c)}}{x} \right) \qquad \text{as } x \to \infty.$$

Now consider I_1. Expand f and ϕ around $t = c$, assuming they are sufficiently differentiable:

$$\begin{aligned} \phi(t) &= \phi(c) + \underbrace{\phi'(c)(t-c)}_{=0} + \tfrac{1}{2}\phi''(c)(t-c)^2 + O((t-c)^3), \\ f(t) &= f(c) + O(t-c) \qquad \text{as } t \to c. \end{aligned}$$

Thus

$$\begin{aligned} I_1(x) &= \int_{c-\epsilon}^{c+\epsilon} [f(c) + \text{H.O.T.}] \exp\left[x\phi(c) + \tfrac{1}{2} x\phi''(c)(t-c)^2 + \text{H.O.T.} \right] dt \\ &= f(c) e^{x\phi(c)} \int_{c-\epsilon}^{c+\epsilon} e^{x\phi''(c)(t-c)^2/2} \left[1 + \text{H.O.T.} \right] dt. \end{aligned}$$

The idea is that the higher-order terms (H.O.T.) in powers of $(t-c)$ generate higher-order terms in an asymptotic series in inverse powers of x (see Chapling, Section 3.3.1, and B&O, Section 6.4). For now, we generate the leading-order term:

$$I_1(x) \sim f(c) e^{x\phi(c)} \int_{c-\epsilon}^{c+\epsilon} e^{x\phi''(c)(t-c)^2/2}\, dt \qquad \text{as } x \to \infty.$$

End of lecture 5

Laplace's method: an interior maximum (continued). In case (b), put

$$s = \sqrt{-x\phi''(c)}\,(t-c), \qquad A = \epsilon\sqrt{-x\phi''(c)}.$$

We use A for the quantity denoted by a on the blackboard, to distinguish it from the lower endpoint of the original integral. The leading local approximation becomes

$$I_1(x) \sim \frac{f(c) e^{x\phi(c)}}{\sqrt{-x\phi''(c)}} \int_{-A}^{A} e^{-s^2/2}\, ds.$$

Let $x \to \infty$, keeping $\epsilon > 0$ fixed. Then $A \to \infty$, and

$$\begin{aligned} I_1(x) &\sim \frac{f(c)e^{x\phi(c)}}{\sqrt{-x\phi''(c)}} \int_{-\infty}^{\infty} e^{-s^2/2}\, ds \\ &= f(c)e^{x\phi(c)} \left(\frac{2\pi}{-x\phi''(c)}\right)^{1/2}. \end{aligned}$$

Since $I_2 \ll I_1$ as $x \to \infty$, we obtain

$$\boxed{I(x) \sim f(c)e^{x\phi(c)} \left(\frac{2\pi}{-x\phi''(c)}\right)^{1/2}} \qquad \text{as } x \to \infty.$$

Error from extending the Gaussian limits. Recall that

$$\int_z^{\infty} e^{-t^2/2}\, dt \sim \frac{1}{z} e^{-z^2/2} \qquad \text{as } z \to \infty.$$

The blackboard records the error from replacing A by ∞ as

$$\text{error} = O\Big(x^{-1/2} e^{-\alpha x^2}\Big) \qquad \text{for some } \alpha > 0.$$

Transcription note (photograph 2): the displayed error is retained as written. With fixed ϵ, however, $A^2 = -\epsilon^2 x\phi''(c)$ is proportional to x. The omitted Gaussian tails, before multiplication by the outer prefactor, therefore have size $O(x^{-1/2}e^{-\alpha x})$, where $\alpha = -\epsilon^2\phi''(c)/2 > 0$. This concerns only the extension of the Gaussian limits; the higher-order Taylor terms give separate corrections.

For an alternative derivation, see Miller, Sections 3.3–3.4. For higher-order terms, see Chapling, Section 3.3.1, and B&O, Section 6.4.

Example 1. Consider

$$I(x) = \int_{-\infty}^{\infty} e^{-xt^2} e^{at}\, dt, \qquad x \to \infty,$$

where a is fixed. Take $f(t) = e^{at}$ and $\phi(t) = -t^2$. Then

$$c = 0, \quad \phi'(0) = 0, \quad \phi''(0) = -2, \quad f(0) = 1, \quad \phi(0) = 0.$$

Hence

$$I(x) \sim \sqrt{\frac{\pi}{x}}.$$

Indeed, completing the square gives the exact value

$$I(x) = e^{a^2/(4x)} \sqrt{\frac{\pi}{x}}.$$

Example 2. The integral written on the blackboard is

$$K_0(x) = \int_0^{\infty} e^{-t - x^2/(4t)}\, dt, \qquad x \to \infty.$$

Transcription note (photographs 5–6): both the symbol $K_0(x)$ and the measure dt are retained from the blackboard. This definition should be checked against the usual special-function notation; the calculation below applies to the integral displayed here.

One might try $f(t) = e^{-t}$ and $\phi(t) = -1/(4t)$, with large parameter x^2. But $\phi'(t) > 0$ for all $t > 0$, so this does not give an interior maximum of ϕ. Instead, consider the whole exponent

$$\psi(t) = -t - \frac{x^2}{4t}, \qquad \psi'(t) = -1 + \frac{x^2}{4t^2}.$$

Its maximum occurs at $t_{\max} = x/2$. This suggests rescaling by

$$xu = 2t, \qquad x\,du = 2\,dt.$$

Then

$$K_0(x) = \frac{x}{2}\int_0^\infty e^{x\phi(u)}\,du, \qquad \phi(u) = -\frac{u}{2} - \frac{1}{2u}.$$

Now

$$\phi'(u) = -\frac{1}{2} + \frac{1}{2u^2} = 0 \quad \Longrightarrow \quad u = 1 = c, \qquad \phi(1) = -1, \quad \phi''(1) = -1.$$

Therefore, for the integral as written,

$$K_0(x) \sim \frac{x}{2}e^{-x}\sqrt{\frac{2\pi}{x}} = \frac{\sqrt{2\pi x}}{2}e^{-x} \qquad \text{as } x \to \infty.$$

(c) Equal absolute maxima at an interior point and an endpoint. Suppose

$$\phi(b) = \phi(c) = \max_{t\in[a,b]} \phi(t), \qquad a < c < b,$$

with

$$\phi'(c) = 0, \quad \phi''(c) < 0, \quad \phi'(b) \neq 0, \qquad f(c) \neq 0, \quad f(b) \neq 0.$$

Figure 2 shows the situation. The nonstationary endpoint contributes a term of order $e^{x\phi(b)}/x$, as in case (a), whereas the interior maximum contributes a term of order $e^{x\phi(c)}/\sqrt{x}$, as in case (b). Since

$$\frac{e^{x\phi(b)}/x}{e^{x\phi(c)}/\sqrt{x}} = x^{-1/2} \to 0,$$

the interior point dominates the integral.

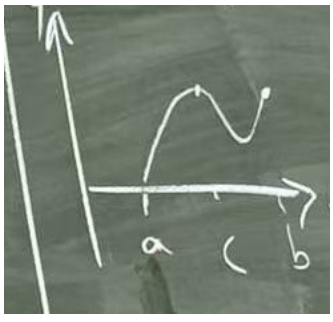


Figure 2: Equal maxima at an interior stationary point c and a nonstationary endpoint b. Source: direct crop of supplied blackboard photograph 6, archived as `source-06.jpg`.

(d) Several interior local maxima. Suppose the interior local maxima c_i, $i = 1, \ldots, n$, satisfy

$$\phi'(c_i) = 0, \qquad \phi''(c_i) < 0, \qquad f(c_i) \neq 0.$$

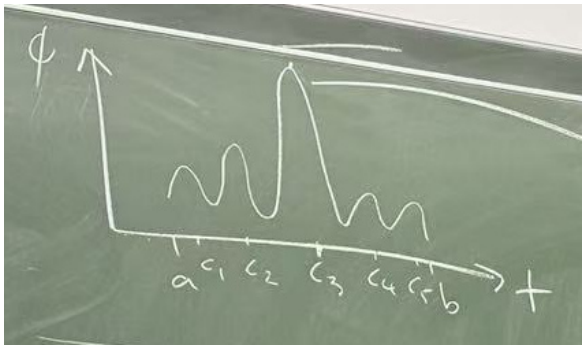
Only the absolute maxima matter at leading order. For instance, if $\phi(c_3) > \phi(c_i)$ for $i = 1, 2, 4, 5$, then

$$\frac{e^{x\phi(c_3)}}{\sqrt{x}} \gg \frac{e^{x\phi(c_i)}}{\sqrt{x}} \qquad \text{as } x \to \infty.$$

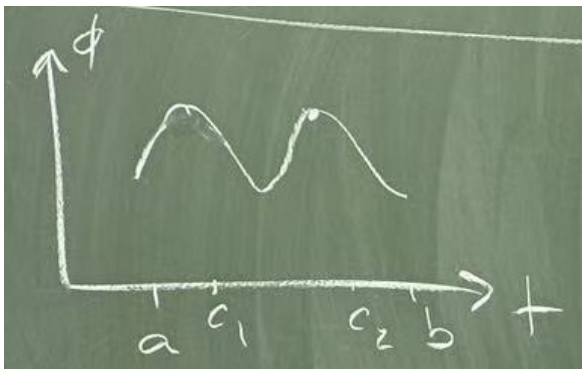
This is illustrated in Figure 3(a). If instead two maxima have equal height, $\phi(c_1) = \phi(c_2) = \max \phi$, both contributions must be included:

$$I(x) \sim e^{x\phi(c_1)} \sqrt{\frac{2\pi}{-x\phi''(c_1)}} + e^{x\phi(c_2)} \sqrt{\frac{2\pi}{-x\phi''(c_2)}}.$$

Transcription note (photograph 8): this is the formula as written on the blackboard. For general f, the two terms must also be multiplied by $f(c_1)$ and $f(c_2)$, respectively; the displayed formula applies when $f \equiv 1$. If the leading coefficients cancel, further terms are needed before asserting an asymptotic equivalence.



(a) One highest maximum.



(b) Two equal maxima.

Figure 3: Several interior maxima: lower maxima are exponentially smaller, while equal absolute maxima contribute together. Source: direct crops of supplied blackboard photograph 8, archived as `source-08.jpg`.

(e) A stationary endpoint maximum. Suppose the endpoint b is a local and absolute maximum, with

$$\phi'(b) = 0, \qquad \phi''(b) < 0, \qquad f(b) \neq 0.$$

Proceed as in case (b), but the local integral now covers only one side of the maximum:

$$\begin{aligned} I_1(x) &= \int_{b-\epsilon}^{b} f(t) e^{x\phi(t)}\, dt \\ &\sim \frac{f(b) e^{x\phi(b)}}{\sqrt{-x\phi''(b)}} \int_{-\infty}^{0} e^{-s^2/2}\, ds \\ &= \frac{1}{2} f(b) e^{x\phi(b)} \sqrt{\frac{2\pi}{-x\phi''(b)}} \qquad \text{as } x \to \infty. \end{aligned}$$

Thus a stationary endpoint gives half the corresponding full Gaussian contribution. The same conclusion holds at a stationary left endpoint.

Example. Consider

$$I(x) = \int_0^{3\pi/2} e^{x \sin^2 t}\, dt, \qquad f(t) = 1, \quad \phi(t) = \sin^2 t.$$

The absolute maxima are at $t = \pi/2$ (interior) and $t = 3\pi/2$ (endpoint), as shown in Figure 4. At both points,

$$\phi = 1, \qquad \phi' = 0, \qquad \phi'' = 2\cos(2t) = -2.$$

Hence

$$\begin{aligned} I(x) &\sim \frac{e^x}{\sqrt{2x}}\sqrt{2\pi} + \frac{1}{2}\frac{e^x}{\sqrt{2x}}\sqrt{2\pi} \\ &= \frac{3}{2}\sqrt{\frac{\pi}{x}}\,e^x \qquad \text{as } x \to \infty. \end{aligned}$$



Figure 4: The phase $\phi(t) = \sin^2 t$ on $[0, 3\pi/2]$, with one interior maximum and one stationary endpoint maximum. Source: direct crop of supplied blackboard photograph 9, archived as `source-09.jpg`.

(f) A higher-order interior maximum. Suppose c is the dominant interior maximum and

$$\phi'(c) = \phi''(c) = \cdots = \phi^{(p-1)}(c) = 0, \qquad \phi^{(p)}(c) < 0,$$

where p is an even positive integer, and $f(c) \neq 0$. Proceed as in case (b), now using

$$\phi(t) = \phi(c) + \frac{\phi^{(p)}(c)}{p!}(t-c)^p + \cdots.$$

Then

$$I(x) \sim f(c)e^{x\phi(c)} \int_{c-\epsilon}^{c+\epsilon} \exp\left[\frac{x\phi^{(p)}(c)}{p!}(t-c)^p\right] dt.$$

Use the rescaling

$$s = \left[\frac{-x\phi^{(p)}(c)}{p!}\right]^{1/p} (t-c).$$

As $x \to \infty$, this gives

$$I(x) \sim f(c)e^{x\phi(c)} \left[\frac{p!}{-x\phi^{(p)}(c)}\right]^{1/p} \int_{-\infty}^{\infty} e^{-s^p}\, ds.$$

To evaluate the remaining integral, put $u = s^p$ on $s \geq 0$:

$$\begin{aligned} \int_0^\infty e^{-s^p}\, ds &= \frac{1}{p}\int_0^\infty e^{-u}u^{1/p-1}\, du \\ &= \frac{1}{p}\Gamma\left(\frac{1}{p}\right) = \Gamma\left(1+\frac{1}{p}\right). \end{aligned}$$

Since p is even, the full integral is twice this value. Therefore

$$\boxed{I(x) \sim 2f(c)e^{x\phi(c)} \left[\frac{p!}{-x\phi^{(p)}(c)}\right]^{1/p} \Gamma\left(1+\frac{1}{p}\right)} \qquad \text{as } x \to \infty.$$

In particular, the algebraic factor is $x^{-1/p}$, so the contribution is proportional to $x^{-1/p}e^{x\phi(c)}$.

End of lecture 6